Importance of Numerical Methods

1. **Solving Complex Problems**
    - Many mathematical equations (especially in science and engineering) cannot be solved exactly using analytical methods. Numerical methods provide approaches to find approximate solutions for equations like differential equations, integrals, and large systems of equations.[1] [2]
2. **Handling Large-Scale Computations**
    - In modern applications, engineers and scientists deal with massive datasets and complex models. Numerical methods enable computers to efficiently process and solve high-dimensional problems that are impractical to solve by hand.[2] [1]
3. **Simulation and Prediction**
    - Numerical methods are used to model and simulate real-world phenomena (e.g., climate modeling, structural analysis, population growth). This allows experts to predict future behavior, optimize designs, and study situations before actual implementation.[3] [4] [2]
4. **Dealing with Uncertainty**
    - Many practical problems include uncertain data or parameters. Numerical techniques support sensitivity analysis and help assess the impact of varying these parameters on system outcomes, supporting reliable decision-making.[1]
5. **Optimization**
    - In design, finance, machine learning, and research, finding optimal solutions is often required. Numerical methods such as iterative algorithms help find the best value for specific objectives, even when exact solutions are unavailable.[1]
6. **Improving Efficiency**
    - Algorithms designed in numerical methods are structured for fast and accurate computation. This makes them essential for time-sensitive tasks and complex simulations run on computers.[5] [1]
7. **Ensuring Accuracy and Reliability**
    - Numerical methods include error analysis and convergence checks to ensure the results are both close to the real values and reliable for practical applications.[2] [5]
8. **Wide Applicability**
    - These methods are fundamental in fields such as engineering, physics, finance, computer science, environmental studies, and medicine for tasks like data fitting, system modeling, imaging, and optimization.[4] [3] [5] [2]

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Exam Tip:

For short questions, focus on the ability of numerical methods to solve otherwise unsolvable problems, handle large computations, and enable simulation/optimization. For long questions, elaborate each point with examples (like weather prediction, FEA in engineering, and stock market modeling in finance).

This format covers both the technical purpose and practical application of numerical methods, suitable for concise and detailed exam answers.

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1. https://onlinenotesnepal.com/importance-of-numerical-methods
2. https://math.libretexts.org/Workbench/Numerical_Methods_with_Applications_(Kaw)/1:_Introduction/1.01:_Introduction_to_Numerical_Methods
3. https://www.alooba.com/skills/concepts/computational-modeling-567/numerical-methods/
4. https://www.slideshare.net/slideshow/importance-of-numerical-methods-in-csepptx/251563924
5. https://byjus.com/maths/numerical-analysis/

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